



7TH GRADE LIFE SCIENCE — CHAPTER 4 REVIEW

Ecology & Ecosystems • S7L4

Parts of an Ecosystem

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An **ecosystem** = all the living + non-living things in an area and how they interact.

Biotic (living)

Plants, animals, fungi, bacteria — anything alive or once alive.

Abiotic (non-living)

Sunlight, water, air, soil, temperature, rocks.

 **Levels:** organism → **population** (same species in an area) → **community** (all populations) → **ecosystem** (adds the non-living parts).

Roles Organisms Play

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Role	How it gets energy	Examples
Producer	Makes its own food (photosynthesis)	Plants, algae
Consumer	Eats other organisms	Rabbits, foxes, humans
Decomposer	Breaks down dead things, returns nutrients	Fungi, bacteria

Consumer types: **Herbivore** (plants only) • **Carnivore** (animals only) • **Omnivore** (both).

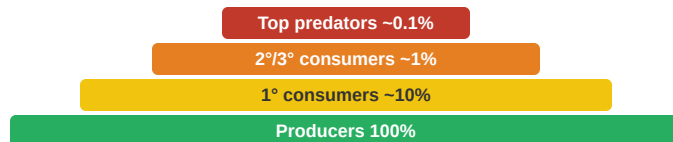
 Without decomposers, dead matter would pile up and nutrients would stay locked away.

Feeding Relationships & Energy Flow

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A **food chain** = one path of energy. A **food web** = all the connected chains.

 Arrows point **from food** → **to the eater** (the way energy flows). Grass → Rabbit.



Energy pyramid — energy shrinks going up.

 **10% Rule:** only ~10% of energy passes to the next level; the rest is lost as heat. That's why there are far more producers than top predators.

Cycling of Matter

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Energy **flows one way** (sun → producers → consumers) and is lost as heat. Matter **cycles** and is reused forever.

Cycle	How it moves
Water cycle	Evaporation → condensation → precipitation → runs back
Carbon cycle	Plants take in CO ₂ ; animals eat & breathe out CO ₂ ; decomposers return carbon

Limiting Factors

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A **limiting factor** keeps a population from growing too large, keeping ecosystems in balance: **resources** (food/water/space), **disease**, **climate** (drought, flood, extreme temps), and **human activity** (pollution, building, overhunting).

 Wolves reintroduced to Yellowstone (1995) controlled elk → trees grew back → beavers & birds returned. One species can reshape a whole ecosystem.

Earth's Biomes & Aquatic Ecosystems

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A **biome** is a large region with its own climate, plants, & animals. The 7 land biomes: tropical rainforest, savanna, temperate forest, desert, grassland, taiga, tundra.

Aquatic type	Description
Freshwater	Low salt — lakes, ponds, rivers, streams
Estuary	Where a freshwater river meets the salty ocean; very rich in life
Marine	Salt water — oceans & coral reefs, the largest ecosystems

Quick Check — Ecology

A. Biotic or abiotic?

1. Sunlight ____
2. A frog ____
3. Soil ____
4. Mushroom ____

B. If grass has 10,000 units of energy, about how much reaches a fox (2 levels up)? ____

C. Producer, consumer, or decomposer? A mushroom on a log → ____

I Can... (Self-Check)

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- | | |
|--|---|
| ☐ Tell biotic from abiotic factors | ☐ Apply the 10% rule on an energy pyramid |
| ☐ Order organism → population → community → ecosystem | ☐ Explain how matter cycles vs. energy flows |
| ☐ Name the roles: producer/consumer/decomposer | ☐ List limiting factors on a population |
| ☐ Read food-web arrows correctly | ☐ Tell apart the major biomes |

 Based on Georgia Standards of Excellence: S7L4 — You're ready for the unit test! 